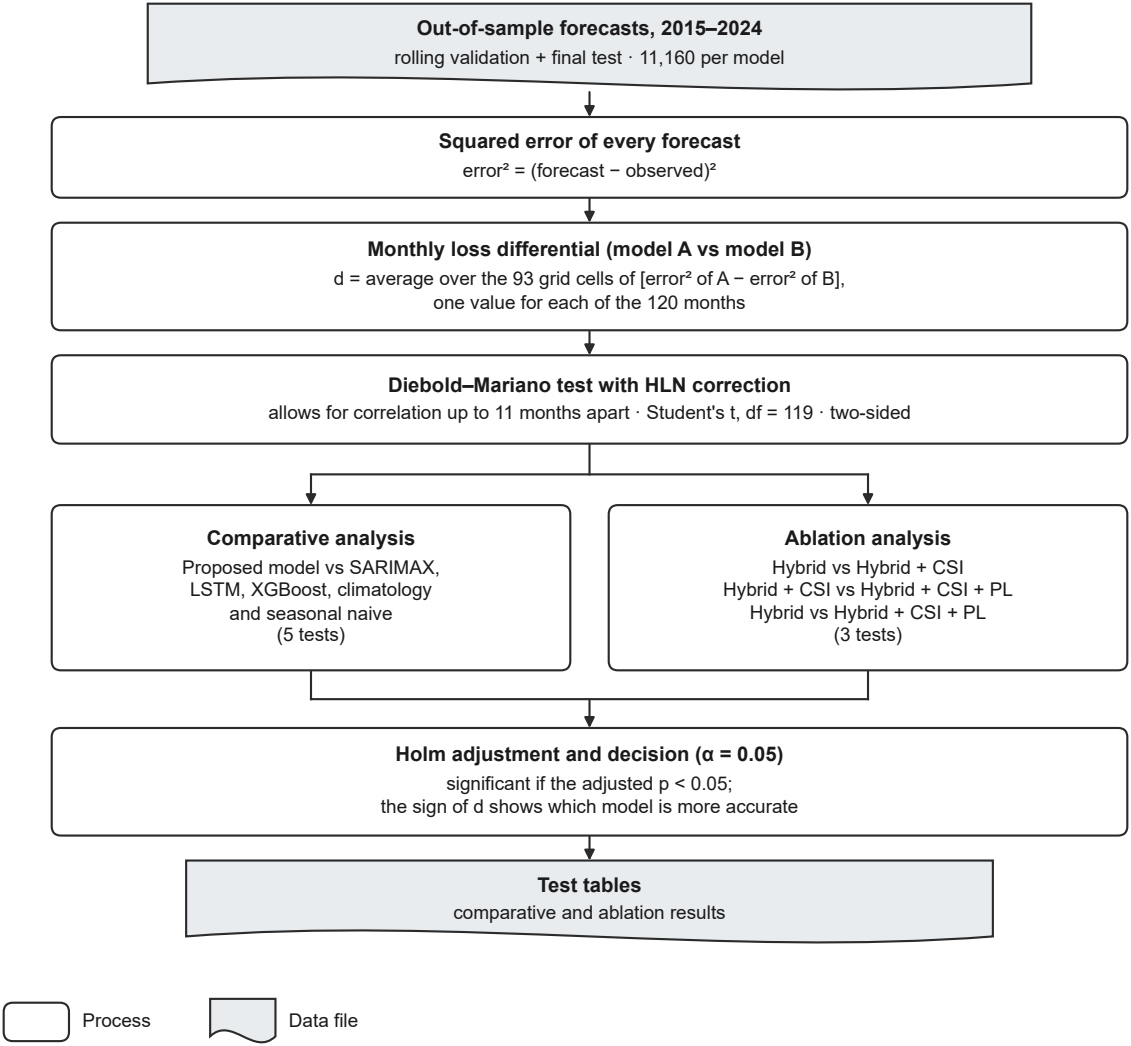

Out-of-sample forecasts, 2015–2024
rolling validation + final test · 11,160 per model
Squared error of every forecast
error² = (forecast − observed)²
Monthly loss differential (model A vs model B)
d = average over the 93 grid cells of [error² of A − error² of B],
one value for each of the 120 months
Diebold–Mariano test with HLN correction
allows for correlation up to 11 months apart · Student's t, df = 119 · two-sided
Comparative analysis
Proposed model vs SARIMAX,
LSTM, XGBoost, climatology
and seasonal naive
(5 tests)
Ablation analysis
Hybrid vs Hybrid + CSI
Hybrid + CSI vs Hybrid + CSI + PL
Hybrid vs Hybrid + CSI + PL
(3 tests)
Holm adjustment and decision (α = 0.05)
significant if the adjusted p < 0.05;
the sign of d shows which model is more accurate
Test tables
comparative and ablation results
Process
Data file